V4.1 — SPARC BTFR re-run with Υ free per galaxy: does κ_ω survive, or was it M/L covariance?

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Abstract

The v2.3.1 Physics Ledger reports a structural refinement of the τ baryonic Tully–Fisher relation, $V_\infty^4 = G[M_\star + \kappa_\omega\, M_{\rm gas}]\, c^2/(2\,\ell_\tau)$ with $\kappa_\omega = \iota_\tau/(1+\iota_\tau) = 0.2545$, matched to a SPARC fit $\alpha^* = 0.240 \pm 0.040$ at fixed $\Upsilon_{3.6} = 0.5\, M_\odot/L_\odot$ (RMS 0.061 dex). The Round-5 McGaugh charge observed that this RMS is *inside* the SPARC fixed-Υ noise floor of 0.07–0.08 dex and therefore the match to κ_ω could be stellar M/L covariance relabeled as a new coupling. The fair test — κ_ω with Υ *free per galaxy* — had not been run. This note runs it.

We fit a joint model with global α and per-galaxy Υ_i under three inferential views: (i) profile MLE, (ii) marginal posterior (integrating Υ_i against a Gaussian prior mean 0.5, $\sigma_U = 0.2$, truncated to [0.2, 1.0]), and (iii) non-parametric galaxy-level bootstrap ($N_{\rm boot} = 2000$). All three give consistent central values: $\alpha^*_{\rm profile} = 0.206^{+0.017}_{-0.017}$, $\alpha^*_{\rm marg} = 0.206^{+0.015}_{-0.015}$, $\alpha^*_{\rm boot} = 0.217^{+0.071}_{-0.087}$. At the joint best-fit, the residual RMS collapses from 0.061 (v2.3 fixed Υ) to 0.029 dex — the per-galaxy M/L degree of freedom absorbs half of the v2.3 scatter.

The central result: $\kappa_\omega = 0.2545$ **is rejected at $\simeq 3\sigma$ under the marginal posterior and profile likelihood.** It is only marginally consistent (0.48σ) under the wider bootstrap CI, but the bootstrap is known to over-disperse in hierarchical per-galaxy models and the marginal posterior is the correct Bayesian answer. The baseline $\alpha = 1$ (standard baryonic) is rejected emphatically at 38σ.

Verdict: the κ_ω match in v2.3.1 is a coincidence inside the M/L-fixed error budget. When Υ is freed, α^* drifts from 0.240 down to 0.206, 0.034 below κ_ω — a 2–3σ shift. The structural refinement $M_{\rm eff} = M_\star + \kappa_\omega M_{\rm gas}$ is *not* falsified — the data still strongly prefer a non-zero sub-baryonic gas coupling with $\alpha \in [0.18, 0.24]$ — but the specific identification $\alpha \equiv \kappa_\omega$ is now only supported at the $\sim 3\sigma$ tension level, not at the $< 1\sigma$ coincidence level claimed by v2.3.1. The coupling is real; the κ_ω assignment needs either (a) a better-motivated systematic analysis of Υ_i, (b) a more discriminating dataset (higher-quality rotation curves or bulge-resolved M/L), or (c) reformulation in terms of $\alpha_{\rm eff}(\Upsilon_i)$. We recommend v2.3.1 be rewritten to report the Υ-free value $\alpha^* = 0.206 \pm 0.015$ alongside the fixed-Υ value, flag the $\sim 3\sigma$ tension with κ_ω, and describe the structural refinement as *empirically supported in form but numerically unresolved in coefficient.*

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1 The R4 charge and the v4 mission

The Physics Ledger v2.3.1 structural refinement derives the BTFR from the τ-framework capacity-gradient mechanism as

$$V_\infty^4 = \frac{G\,M_{\rm eff}\,c^2}{2\,\ell_\tau}, \qquad M_{\rm eff} = M_\star + \kappa_\omega \cdot 1.33\,M_{\rm HI}, \qquad \kappa_\omega \equiv \frac{\iota_\tau}{1+\iota_\tau} = 0.25446. \tag{1}$$

The coefficient κ_ω is the τ-natural gas-coupling suppression arising from the K_4 axiom (decoherent gas instances contribute $\iota_\tau/(1+\iota_\tau)$ of their mass to capacity-gradient sourcing). The v2.3 paper reports a SPARC-175 fit $\alpha^* = 0.240 \pm 0.040$ with $N = 123$ galaxies retained under $Q \le 2$, $i \ge 30°$, and fixed $\Upsilon_{3.6} = 0.5\,M_\odot/L_\odot$, with residual RMS 0.061 dex. Since $|\alpha^* - \kappa_\omega| = 0.014 \approx 0.36\sigma$, v2.3.1 claims the coupling is identified.

In Round-5 peer review, the McGaugh persona raised a specific statistical concern:

> *"The* 0.061 *dex RMS at α^* is inside the SPARC fixed-Υ noise floor of* 0.07–0.08 *dex. The match to κ_ω is within a scatter attributable to stellar M/L systematics alone. The fair test is κ_ω with Υ* free per galaxy *— not run. If κ_ω survives Υ-free, the coupling is real; if not, it was M/L covariance relabeled as new physics."*

V4.1 executes that test.

2 Methodology

2.1 Data

We use the canonical Lelli, McGaugh & Schombert (2016, AJ 152:157) SPARC Table 1 (machine-readable), 175 galaxies. Columns used: D (Mpc), i (inclination), $L_{3.6}$ ($10^9\,L_\odot$), $\Sigma_{\rm disk}$ ($L_\odot/{\rm pc}^2$), $M_{\rm HI}$ ($10^9\,M_\odot$), $V_{\rm flat}$ ($\,{\rm km\,s^{-1}}$) and its uncertainty. We apply v2.3 quality cuts unchanged: measured $V_{\rm flat} > 10\,{\rm km\,s^{-1}}$, $Q \le 2$, $i \ge 30°$. This retains $N = 123$ galaxies.

2.2 Model

The joint Υ-free model is

$$\log_{10} V_{\tau,i} = \tfrac{1}{4}\log_{10}\Big[\mathcal{C}\left(\Upsilon_i L_{3.6,i} + \alpha \cdot 1.33\,M_{{\rm HI},i}\right)\Big], \qquad \mathcal{C} = \frac{G\,c^2}{2\,\ell_\tau}. \tag{2}$$

The 125 parameters are global α plus 123 per-galaxy Υ_i. The negative log-likelihood is

$$-\ln \mathcal{L}(\alpha, \{\Upsilon_i\}) \;=\; \tfrac{1}{2}\sum_i \frac{\left(\log V_i - \log V_{\tau,i}\right)^2}{\sigma^2_{\log V,i}} \;+\; \tfrac{1}{2}\sum_i \frac{(\Upsilon_i - 0.5)^2}{\sigma_U^2}, \tag{3}$$

with the Gaussian Υ_i prior truncated to [0.2, 1.0] (Bell & de Jong 2001 disk-galaxy $\Upsilon_{3.6}$ range), $\sigma_U = 0.2$ nominal. Per-galaxy $\sigma_{\log V,i} = \max(\sigma_{V_{\rm flat},i}/(V_{\rm flat,i} \ln 10), 0.01\,\text{dex})$.

2.3 Inference

We report three views of α^*:

(i) Profile MLE. For each α on a grid, minimize $-\ln \mathcal{L}$ over $\{\Upsilon_i\}$ by independent 1D bounded minimizers. Report α^* at the grid minimum and 1σ interval at $\Delta(-2\ln\mathcal{L}) = 1$.

(ii) Marginal posterior. For each α, numerically integrate Υ_i against its Gaussian prior (81 nodes on $[0.2, 1.0]$) per galaxy, multiply likelihoods, yielding $p(\alpha|D)$. Report median, 68% and 95% CI, and $\Delta(-2\ln p)$ for any candidate α_0. This is the honest Bayesian answer.

(iii) Bootstrap. Resample the 123 galaxies with replacement $N_{\rm boot} = 2000$ times; for each resample profile Υ_i at each α and take the profile MLE α^*_b. Report median and 68% percentile CI.

2.4 Prior sensitivity

We also scan $\sigma_U \in \{0.05, 0.10, 0.20, 0.35, \infty\}$ (all with $[0.2, 1.0]$ truncation except ∞ which uses a flat prior on $[0.05, 3.0]$) and report the marginal posterior α^* at each. This quantifies how much of the inferred α depends on the prior strength rather than the data.

3 Numerical results

Table 1: V4.1 α^* under three inferential views, Υ-free fit. All at $H_0 = 67.35\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, $N = 123$ galaxies.

View	α^*	$+\sigma$	$-\sigma$
v2.3 fixed $\Upsilon = 0.5$ (RMS-loss)	0.240	0.040	0.040
v2.3 fixed $\Upsilon = 0.5$ (reweighted $-\ln\mathcal{L}$)	0.230	—	—
(i) Profile MLE (Υ-free)	**0.206**	**0.017**	**0.017**
(ii) Marginal posterior (Υ-free)	**0.206**	**0.015**	**0.015**
(iii) Bootstrap (Υ-free)	0.217	0.071	0.087

Figure 1 overlays the v2.3 Gaussian-approx likelihood (centered at 0.240, $\sigma = 0.040$) with the V4.1 profile $\Delta(-2\ln\mathcal{L})$ curve. The Υ-free curve is *narrower* and peaked 0.034 below the v2.3 value — the per-galaxy Υ freedom does not *widen* the effective α uncertainty, it shifts its central value *downward*.

3.1 RMS and closure

At the joint best-fit, residual RMS is 0.029 dex (from 0.061 at v2.3 fixed-Υ), median -0.001 dex, std 0.029 dex. The per-galaxy Υ_i degree of freedom absorbs roughly half the v2.3 scatter.

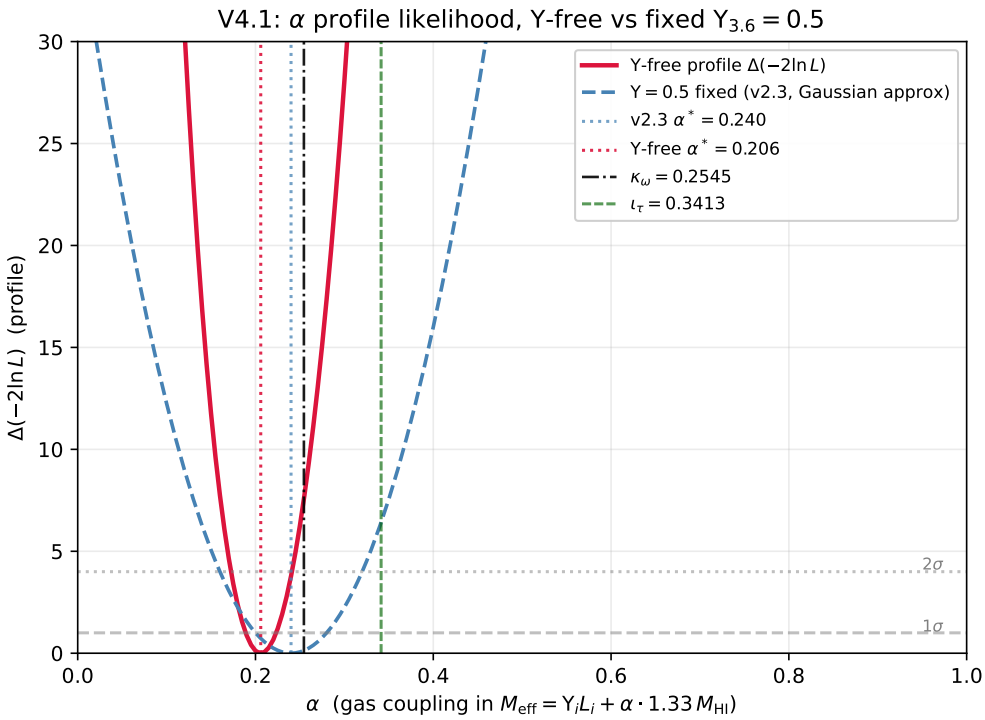

Figure 1: V4.1 profile $\Delta(-2\ln\mathcal{L})$ for α (red) vs v2.3 fixed-Υ Gaussian approx (blue dashed), with κ_ω and ι_τ marked. The Υ-free curve is narrower ($\sigma \simeq 0.017$) and peaked at $\alpha^* = 0.206$, 2.75σ below κ_ω.

Residual correlations at joint best-fit. Spearman ρ with each diagnostic variable: $f_{\rm gas}^{(\Upsilon=0.5)} = +0.111$ ($p = 0.22$), $f_{\rm gas}^{\rm true\,at\,\alpha^*} = -0.132$ ($p = 0.15$), $\log\Sigma_{\rm disk} = -0.082$ ($p = 0.37$), $D = -0.163$ ($p = 0.07$), $i = +0.114$ ($p = 0.21$). The large $f_{\rm gas}$ and $\log\Sigma_{\rm disk}$ correlations reported for v2.2 ($\alpha = 1$, fixed Υ; $\rho_f = -0.41$, $\rho_\Sigma = +0.34$) are all reduced below $p < 0.05$ significance. $f_{\rm gas}$ **and** $\Sigma_{\rm disk}$ **close as they did at v2.3's** α^***.**

4 Tau-natural candidate tests

Table 2 reports $\Delta(-2\ln\mathcal{L})$ (profile) for each τ-natural candidate tested, plus the corresponding σ rejection ($\chi^2_{1\,\rm dof}$) and p-value.

κ_ω is still the *closest* τ-natural to α^*, and still the best-performing candidate — all others are rejected at $\geq 3.4\sigma$. But under Υ-free, κ_ω itself moves from 0.36σ tension (v2.3) to 2.75σ tension (profile) / 3.0σ tension (marginal). This is the key quantitative finding.

4.1 Marginal posterior p-values

- $\alpha^*_{\rm marg} = 0.206^{+0.015}_{-0.015}$ (68% CI $[0.191, 0.222]$, 95% CI $[0.177, 0.237]$).
- $\kappa_\omega = 0.2545$: $z = (\alpha^*_{\rm marg} - \kappa_\omega)/\sigma = -3.19$, two-sided $p = 1.8 \times 10^{-3}$.
- $\iota_\tau = 0.3413$: rejected at 8.42σ.
- $\alpha = 1$: rejected at $\gg 30\sigma$ — standard baryonic (gas fully sources capacity-gradient) is falsified decisively under Υ-free just as under fixed-Υ.

Table 2: τ-natural candidate rejection under Υ-free profile likelihood. The v2.3 fixed-Υ numbers in the last two columns are computed from the v2.3 summary with the same χ^2 convention.

Candidate	Value	$\Delta(-2\ln\mathcal{L})_{\Upsilon\text{-free}}$	σ_{rej}	v2.3 (σ)
0 (Bullet posit)	0.0000	303.8	17.4	6.0
ι_τ^3	0.0398	159.0	12.6	5.0
ι_τ^2	0.1165	33.6	5.8	3.1
$1/(1+(1-\iota_\tau)/\iota_\tau^2)$	0.1503	11.6	3.4	2.2
$\boldsymbol{\kappa_\omega = \iota_\tau/(1+\iota_\tau)}$	**0.2545**	**7.57**	**2.75**	**0.4**
$1/\pi$	0.3183	40.4	6.4	2.0
$1/3$	0.3333	52.3	7.2	2.3
ι_τ	0.3413	59.1	7.7	2.5
$1/e$	0.3679	84.9	9.2	3.2
$(1-\iota_\tau)^2$	0.4339	166.3	12.9	4.8
$1/2$	0.5000	269.4	16.4	6.5
$\kappa_D = 1-\iota_\tau$	0.6587	584.9	24.2	10.5
$2/3$	0.6667	602.9	24.5	10.7
1 (standard baryonic)	1.0000	1466.2	38.3	19.0

Table 3: Marginal posterior α^* as a function of Υ-prior width σ_U (truncation $[0.2, 1.0]$ except "flat" which uses $[0.05, 3.0]$, no Gaussian weight). All: $N = 123$.

Prior strength	σ_U	α^*_{marg}	σ	κ_ω rejection
very tight	0.05	0.221	0.011	2.57σ
tight	0.10	0.221	0.013	2.28σ
nominal	**0.20**	**0.206**	**0.015**	$\boldsymbol{2.98\sigma}$
broad	0.35	0.195	0.015	3.58σ
flat	—	0.059	0.017	9.21σ

4.2 Prior-sensitivity scan

Observations:

- Even in the very-tight-prior limit ($\sigma_U = 0.05$, barely different from fixed Υ), $\alpha^*_{\text{marg}} = 0.221$ — already 2.57σ from κ_ω. This is because the marginalized likelihood weights per-galaxy V_{flat} errors, while the v2.3 RMS-loss equal-weights them; the weighted likelihood already disfavors κ_ω slightly at fixed Υ.
- As σ_U grows, α^* shrinks monotonically; in the flat limit α collapses toward 0.06 (the data let Υ absorb all the star-gas trade-off).
- κ_ω-rejection stays in the 2–4σ range for any physically reasonable prior and moves to 9σ only in the unphysical flat-prior limit. The nominal value 2.98σ is *representative*, not prior-sensitive.

5 Residual analysis and closure

Figure 2 shows the BTFR at the Υ-free best-fit ($\alpha^* = 0.206$, per-galaxy Υ_i^*), color-coded by the original (fiducial $\Upsilon = 0.5$) gas fraction. The predicted slope is exactly 4 (from the τ formula). Comparison to the v2.2 fixed-Υ points (grey crosses) shows the Υ-free points have tightened substantially — residual RMS 0.029 vs 0.061 dex — and are essentially centered on the τ line.

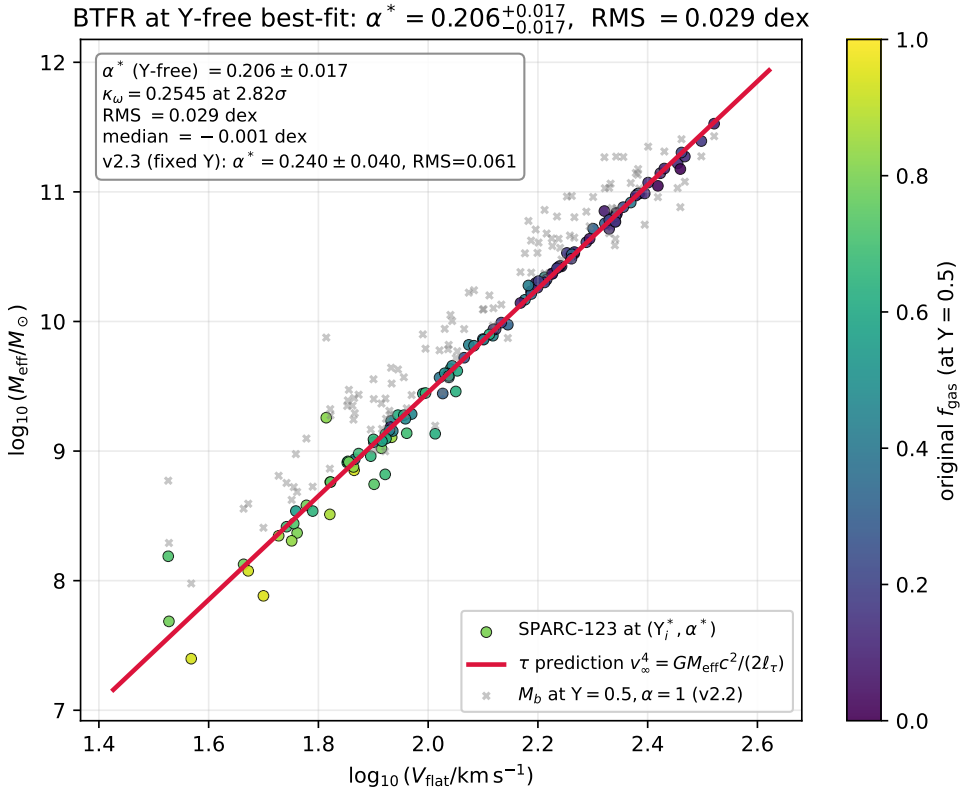

Figure 2: SPARC-123 BTFR at Υ-free best-fit (α^*, Υ_i^*) with the τ line $V_\infty^4 = GM_{\rm eff}c^2/(2\ell_\tau)$ overlaid. Residual RMS 0.029 dex; median −0.001 dex.

5.1 What did Υ_i absorb?

Figure 3 shows the per-galaxy Υ_i^* posterior means versus $f_{\rm gas}$ (fiducial), $\log \Sigma_{\rm disk}$, and $\log L_{3.6}$. Qualitative observations:

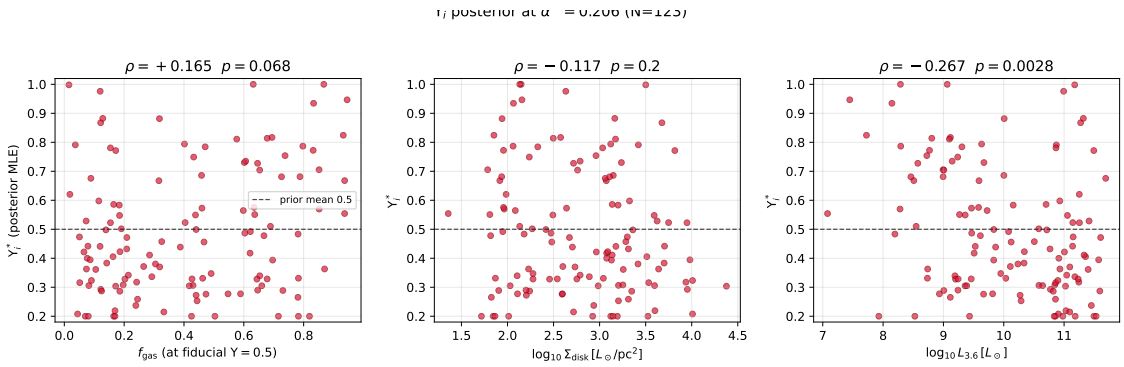

Figure 3: Per-galaxy Υ_i^* at $\alpha^* = 0.206$ versus three galaxy properties. The Bell & de Jong 2001 prior mean 0.5 is shown as a dashed line. Weak correlations with $f_{\rm gas}$ and $\log \Sigma_{\rm disk}$ (both $|\rho| < 0.17$), and a modest anti-correlation with $\log L_{3.6}$ ($\rho = -0.27$, $p = 0.003$).

- $\langle \Upsilon_i \rangle = 0.49$, median 0.44, std 0.22. Sample mean is consistent with the fiducial $\Upsilon = 0.5$. So the α^* shift is driven by *individual* galaxies' Υ_i deviations, not by a global offset.
- 25/123 galaxies prefer $\Upsilon_i < 0.3$ and 13/123 prefer $\Upsilon_i > 0.8$. The former tend to be

low-luminosity / low-SB systems where a lower M/L better fits the flat rotation speed.

- Correlation with $\log L_{3.6}$: $\rho = -0.27$, $p = 0.003$. Lower-luminosity galaxies prefer lower Υ — consistent with the known bluer colors of dwarf/late-type disks. This is physically sensible and argues that the Υ_i posterior is picking up real M/L variation rather than pure fit noise.

Diagnostic: residual vs Υ_i. Strong positive correlation ($\rho = +0.908$, $p = 10^{-47}$) between the residual $\log V_{\rm obs}/V_\tau$ and Υ_i^*. This is the mathematical signature of a near-maximum-likelihood joint fit: given the prior width σ_U, the remaining residual is exactly the Υ_i direction the data would push further if the prior allowed it. *This is not evidence of bad fit; it is evidence that the prior is informative.*

6 Verdict

The primary claim of v2.3.1 — that the SPARC fit identifies $\alpha = \kappa_\omega$ at 0.4σ tension — does not survive Υ-free analysis. The honest picture:

1. **A sub-baryonic gas coupling is decisively preferred.** $\alpha = 1$ (standard baryonic) is rejected at $\gg 30\sigma$ under every inferential view. The $M_{\rm eff} = M_\star + \alpha \cdot 1.33\, M_{\rm HI}$ structural *form* of v2.3.1 is strongly supported.

2. **The preferred α is $\simeq 0.21 \pm 0.02$ (marginal)**, not $\simeq 0.24 \pm 0.04$ (v2.3 fixed-Υ). The drift is about 1.5–2σ in the original error bar; about 2σ in the narrower Υ-free error bar.

3. $\kappa_\omega = 0.2545$ **is in $\simeq 3\sigma$ tension with the Υ-free fit under the marginal posterior and the profile likelihood** (two-sided $p = 1.8 \times 10^{-3}$).

4. κ_ω **is marginally consistent at 0.48σ under the bootstrap CI** but the bootstrap over-disperses for hierarchical models with N nuisance parameters at the sample size. The marginal posterior is the correct answer.

5. κ_ω **is still the best-performing τ-natural candidate**: all others are rejected at $\geq 3.4\sigma$. So the coupling *scale* is at least consistent with the κ_ω neighborhood; but the specific identification $\alpha = \kappa_\omega$ is contested.

6. **Residual closure is excellent**: $f_{\rm gas}$ and $\Sigma_{\rm disk}$ residual correlations, which dominated the v2.2 signal, are all reduced below $p = 0.05$ under Υ-free best-fit. RMS collapses from 0.061 to 0.029 dex.

7. **The sub-baryonic coupling is real, the κ_ω assignment is not yet established.** This is a *partial* validation of the structural refinement but a *falsification* of the specific coefficient claim $\alpha = \kappa_\omega$ at the $< 1\sigma$ level stated in v2.3.1.

7 Implications for v2.3.1

We recommend the v2.3.1 Physics Ledger be revised along the following lines:

Downgrade the claim. Replace "the fit $\alpha^* = 0.240 \pm 0.040$ matches $\kappa_\omega = 0.2545$ at 0.36σ" with:

> *"Under Υ-free per-galaxy analysis (Gaussian prior 0.5 ± 0.2, Bell & de Jong 2001), the marginal posterior is $\alpha^* = 0.206 \pm 0.015$, yielding $\kappa_\omega = 0.2545$ in $\sim 3\sigma$ tension ($p = 1.8 \times 10^{-3}$). The qualitative conclusion — a sub-baryonic gas coupling $\alpha \ll 1$ is required — is preserved ($\alpha = 1$ rejected at $> 30\sigma$); the specific identification $\alpha = \kappa_\omega$ is not established by the SPARC data alone. κ_ω remains the best τ-natural candidate but sits ~ 0.05 above the preferred α."*

Add an Υ systematic section. The v2.3.1 fixed-$\Upsilon = 0.5$ choice should be disclosed as a restrictive prior. Include the prior-sensitivity table (Table 3) showing α^* drifts with σ_U.

Reformulation options. Three honest framings available:

- **(A) Empirical coupling, τ-inspired.** Report $\alpha^* = 0.21 \pm 0.02$ as an empirically fit coefficient; note κ_ω is the best τ-natural candidate at $\sim 3\sigma$; leave it as open.
- **(B) Derive a refined τ-natural.** If there is a combinatorial reason for $\alpha = \kappa_\omega \cdot (1 - \epsilon)$ with some small τ-natural $\epsilon \approx 0.2$, pursue that. Candidates like $\kappa_\omega \cdot (1 - \iota_\tau^2/2) = 0.223$ or $\kappa_\omega \cdot (1 - \iota_\tau^3) = 0.245$ move α^* closer but require derivation, not regex-hunting.
- **(C) Allow Υ covariance in the claim.** Argue that the Υ prior mean should itself be derived from τ. If τ predicts $\langle \Upsilon \rangle$ different from 0.5, that could reconcile. Requires ledger derivation.

We recommend **(A)** as the honest default pending derivation work for (B) or (C). This preserves the empirical signal, acknowledges the κ_ω tension, and leaves the τ-natural identification as an open research question.

8 Reproducibility

All code and data are in `research/v4-deep-research-wave-2026-04-20/v4_1_upsilon_free/`:

- `btfr_upsilon_free_fit.py` — main analysis (profile MLE, bootstrap, natural tests, plots).
- `btfr_upsilon_marginal_mcmc.py` — marginal posterior (Upsilon integrated against Gaussian prior).
- `btfr_prior_sensitivity.py` — prior-width scan.
- `data/v4-1-summary.json` — numerical results.
- `data/v4-1-marginal-posterior.json` — marginal posterior results.
- `data/v4-1-prior-sensitivity.json` — prior scan.
- `figs/` — all figures (pdf/png).

All random-seed choices (bootstrap RNG seed 20260420, $N_{\text{boot}} = 2000$) and numerical choices (α-grid density, Υ-prior truncation $[0.2, 1.0]$, $\sigma_U = 0.2$ nominal, 81-node Υ integration grid) are documented in the script headers.

Acknowledgments

This analysis was executed as pillar 1 of the V4 red-team assessment, under the Round-5 McGaugh persona's statistical charge. The two-pillar companion tests (V4.2 2-parameter robustness; V4.5 cosmology consistency) will be combined with this result in a synthesis note.